Sign-Balanced Liquidation Clearing in DeFi Markets

Abstract. We study DeFi liquidation clearing with collateral-only sales and linear price impact. Even without debt-side buy pressure, clearing can be nonmonotone: selling an asset can lower another borrower's debt value and thereby rescue that borrower.

We show that the asset collateral-debt graph controls the complexity boundary. If the graph is bipartite and price impact is linear, a sign transform turns smooth clearing into a subtraction-free clipped-affine system; a Baker-free activation cascade computes an exact clearing in polynomial time for all drift, large impact included. The same sign transform gives a monotone Tarski map, and therefore a CLS = PPAD ∩ PLS upper bound, for separable monotone nonlinear impact curves such as independent AMM pools, where the affine cascade no longer applies. Conversely, non-bipartite linear instances can implement NOT gates and encode SUM/NOT generalized circuits, making constant-approximate clearing PPAD-hard; smooth approximate clearing is in PPAD. In the all-or-nothing model, balanced clearing is polynomial-time solvable while connected non-bipartite clearing existence is NP-complete.

Keywords: DeFi liquidation · Clearing · PPAD · CLS · Tarski fixed points · Financial networks

1 Introduction and Technical Overview

DeFi lending protocols liquidate undercollateralized positions. A position has a collateral asset and a debt asset. If it becomes unhealthy, liquidators repay debt and receive collateral; at the market level, the relevant price impact is that collateral is sold. When positions are large relative to liquidity, these sales move prices, and those price movements affect the health of other positions. A clearing is a self-consistent vector of liquidations and prices: the liquidations implied by prices must be exactly the liquidations whose collateral sales generate those prices.

This paper studies the computational problem created by this feedback. The model is deliberately conservative: liquidations sell collateral, debts are repaid externally, and there is no debt-side market buy pressure. The nonmonotonicity comes solely from an asset playing collateral in one position and debt in another.

A DeFi-native negative feedback. Consider three assets A, B, C and three positions

$$P_1 = (A, B), \qquad P_2 = (B, C), \qquad P_3 = (C, A),$$

where (A, B) denotes collateral A and debt B. If P_1 is liquidated more aggressively, more A is sold and p_A falls. But A is the debt asset of P_3, so the value of

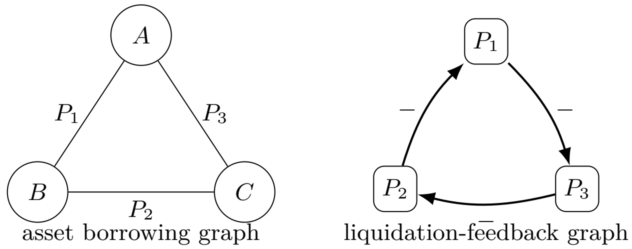

Fig. 1. A three-asset collateral-debt cycle. Selling an asset can lower another position's debt value, creating a negative liquidation dependency.

P_3's debt falls. Thus P_3 becomes healthier and may liquidate less. A liquidation can therefore rescue another borrower whose debt is denominated in the asset being sold. This effect creates an all-negative cycle of liquidation feedback; see Fig. 1. It is the primitive behind both our sign-balance theorem and our hardness gadgets.

Positioning. The novelty is not that clearing can have a Tarski/lattice structure; that is known for several financial clearing and fire-sale models. The contribution is to identify the DeFi-specific sign structure created by collateral-only price impact and debt relief: the collateral-debt graph determines whether this feedback can be globally aligned into a subtraction-free system (polynomial-time for linear impact, CLS for monotone nonlinear impact) or can implement SUM/NOT circuits and PPAD-hard fixed-point structure.

Main results. Let G_A be the asset borrowing graph: vertices are assets and each position contributes the edge between its collateral and debt assets. If G_A is bipartite, flipping one side of the asset prices and complementing liquidation variables whose collateral lies on one side transforms the equations into

$$y = \Pi_{[0,u]}(Ay + c), \qquad A \geq 0.$$

For linear price impact, Section 3.1 gives a polynomial-time exact algorithm for this subtraction-free box fixed-point problem for arbitrary drift, so bipartite linear clearing is in P. The same monotonicity also has a robust Tarski consequence: grid rounding gives a CLS upper bound, and Appendix A extends this to separable monotone nonlinear impact curves, including independent AMM pools. For that genuinely nonlinear sign-balanced class, improving the CLS upper bound to P, or proving CLS-hardness, remains open.

On the hard side, we reduce from generalized circuits using only truncated-SUM and NOT gates. The source circuit is made intrinsically unbalanced: it contains an odd-NOT cycle in the same dependency component as a hard wire. The liquidation gadgets preserve parity, with NOT gates becoming odd collateral-debt paths and SUM gates becoming even paths. Hence the hard instances are non-bipartite because the encoded circuit is sign-frustrated, not because of a disconnected anchor. Smooth approximate clearing is in PPAD, and the all-or-nothing version is polynomial for balanced instances but NP-complete for connected non-bipartite ones.

Thus the paper contributes a DeFi-native collateral-only clearing model and the following map: bipartite linear smooth clearing is exactly polynomial-time solvable; bipartite separable monotone impact lies in CLS, with P vs. CLS-hardness open beyond affine maps; non-bipartite linear smooth clearing is constant-δ PPAD-hard; and all-or-nothing clearing is P for balanced instances but NP-complete for connected non-bipartite ones. The hardness statements are worst-case: small-impact non-bipartite instances can still be contractions.

2 Model and Sign Structure

There are assets $k \in [n]$, prices $p_k \in [0, M_k]$, and positions $i \in [m]$. Position i has collateral asset a_i, debt asset b_i, collateral amount $C_i > 0$, debt amount $D_i > 0$, liquidation threshold $\theta_i \geq 1$, and smoothing width $\varepsilon_i > 0$. We assume $a_i \neq b_i$. The health deficit is

$$h_i(p) = \theta_i D_i p_{b_i} - C_i p_{a_i}. \tag{1}$$

A positive deficit means the position is unhealthy. In the smooth model, liquidation intensity is

$$x_i = \Pi_{[0,1]}\left(\frac{h_i(p)}{\varepsilon_i}\right). \tag{2}$$

Liquidation sells collateral only. Thus the price equation is

$$p_k = \Pi_{[0,M_k]}\left(p_k^0 - s_k \sum_{i:a_i=k} \alpha_i x_i\right), \tag{3}$$

where $s_k > 0$ is the impact slope and $\alpha_i \geq 0$ is the collateral amount sold by a fully liquidated position. When we impose a close-factor constraint, we require $0 \leq \alpha_i \leq \varphi C_i$ for a fixed $\varphi \in (0, 1]$. Debt is repaid externally and does not create a market buy order. The clipping makes the clearing map a continuous self-map of a compact box. The hardness construction below respects the close-factor bound; it adjusts output weights through market-depth parameters s_k, not by selling more collateral than a position contains.

When several positions share a collateral asset, all positive collateral-side couplings factor through the same price and slope, giving a rank-one block; the all-or-nothing gadget sheet uses this constraint explicitly in Lemma 15.

A smooth exact clearing is a pair (p, x) satisfying the price and liquidation equations. A δ-clearing satisfies

$$\max\{\|p - P(x)\|_\infty, \|x - X(p)\|_\infty\} \leq \delta,$$

where P and X denote the right-hand sides of (3) and the liquidation equation. In the all-or-nothing model, $x_i \in \{0, 1\}$; a clearing is a threshold-consistent liquidation set.

All parameters in the input are rational numbers encoded in binary. The computational problems ask for a δ-approximate clearing for a rational approximation parameter δ; in the contraction regimes below, exact or high-precision approximate clearings can be obtained by iteration. In the hardness theorem, δ is a fixed positive constant determined by the source generalized-circuit hardness constant and by the constant error amplification of the gadgets. The price clipping bounds M_k are part of the input; in the reduction they are constants.

We use two graphs. The asset borrowing graph G_A has assets as vertices and an undirected edge $\{a_i, b_i\}$ for each position. This is the graph used in the tractability and hardness statements. The position signed coupling graph G_X, formalized in Definition 1, has positions as vertices, a positive edge between i, j when $a_i = a_j$, and a negative edge when $a_i = b_j$ or $a_j = b_i$. For local intuition we also read it directionally: a positive arc $j \to i$ when $a_j = a_i$, and a negative arc $j \to i$ when $a_j = b_i$. Balance is a property of the undirected graph.

Economic meaning of bipartiteness. Bipartiteness says assets can be separated into two sectors and every borrowing position crosses the sector boundary; it rules out odd cycles in which an asset returns with the opposite collateral/debt role, the source of Fig. 1. The collateral-only price map is competitive on G_A: increasing a debt-asset price can only increase liquidations of positions borrowing that asset, hence weakly decrease their collateral prices. At the liquidation-intensity level, common-collateral co-writers create positive dependencies, while collateral-as-debt reads create negative dependencies.

The algorithmic boundary in the paper is stated on G_A. If G_A is bipartite, one global sign convention orders prices on one side normally and prices on the other side in reverse; the parallel transform on liquidation variables gives the monotone system of Theorem 1. Non-bipartiteness is a worst-case source of hardness rather than an instance-level iff criterion: small-impact non-bipartite instances can still be contractions, while the large-impact constructions in Section 5 encode PPAD-complete fixed points.

3 Sign-Balanced Clearing, Tarski, and Small-Impact Tractability

Theorem 1. *In the collateral-only linear price-impact model with price clipping, if the asset borrowing graph G_A is bipartite, then for every rational $\delta > 0$, smooth δ-clearing reduces in polynomial time to the succinct Tarski fixed-point problem. Hence bipartite smooth δ-clearing belongs to* CLS = PPAD ∩ PLS.

Let (L, R) be a bipartition of G_A. Define transformed price variables

$$q_k = \begin{cases} p_k, & k \in L, \\ M_k - p_k, & k \in R, \end{cases} \tag{4}$$

and transformed liquidation variables

$$z_i = \begin{cases} 1 - x_i, & a_i \in L, \\ x_i, & a_i \in R. \end{cases} \tag{5}$$

Let $y = (q, z)$ and let $[0, u]$ be the corresponding box, with upper bounds M_k for transformed prices and 1 for transformed liquidations.

Lemma 1. *Under the above transform, the clearing equations are equivalent to*

$$y = \Pi_{[0,u]}(Ay + c) \tag{6}$$

for a rational matrix $A \geq 0$ and vector c. Moreover, after ordering price and liquidation variables separately, A has block form

$$A = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix}, \qquad P, Q \geq 0.$$

Proof. There are four cases. First, if $k \in L$, then every position with collateral k has $x_i = 1 - z_i$, and

$$q_k = p_k = \Pi_{[0,M_k]}\left(p_k^0 - \sum_{i:a_i=k} s_k \alpha_i + \sum_{i:a_i=k} s_k \alpha_i z_i\right),$$

which has nonnegative coefficients on the z_i's. Second, if $k \in R$, then $q_k = M_k - p_k$ and $x_i = z_i$ for positions collateralized by k. Using $M - \Pi_{[0,M]}(r) = \Pi_{[0,M]}(M - r)$,

$$q_k = \Pi_{[0,M_k]}\left(M_k - p_k^0 + \sum_{i:a_i=k} s_k \alpha_i z_i\right),$$

again with nonnegative coefficients.

For liquidations, consider $a_i \in L, b_i \in R$. Then $p_{a_i} = q_{a_i}$, $p_{b_i} = M_{b_i} - q_{b_i}$, and $z_i = 1 - x_i$. Since $1 - \Pi_{[0,1]}(r) = \Pi_{[0,1]}(1 - r)$,

$$z_i = \Pi_{[0,1]}\left(1 - \frac{\theta_i D_i M_{b_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i}\right).$$

Finally, if $a_i \in R, b_i \in L$, then $z_i = x_i$, $p_{a_i} = M_{a_i} - q_{a_i}$, and $p_{b_i} = q_{b_i}$, giving

$$z_i = \Pi_{[0,1]}\left(-\frac{C_i M_{a_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i}\right).$$

All displayed coefficients of variables are nonnegative. Price equations depend only on transformed liquidation variables, and liquidation equations depend only on transformed price variables, which gives the block-off-diagonal form.

The map

$$T(y) = \Pi_{[0,u]}(Ay + c)$$

is therefore isotone in the product order. This is the Tarski consequence of sign-balance. For linear impact it will be strengthened to an exact polynomial-time algorithm in Section 3.1; the CLS statement below records the robust monotone fixed-point upper bound that also survives the nonlinear extension in Appendix A.

Proof (Proof of Theorem 1). Fix $\delta > 0$. Choose a rational grid step $\eta \leq \delta$ with polynomially many bits, and let

$$\mathcal{G}_\eta = [0, u] \cap \eta\mathbb{Z}^d$$

with the natural product order. Define a grid map

$$T_\eta(y) = \lfloor T(y) \rfloor_\eta,$$

where the floor is taken coordinate-wise to the largest grid point not exceeding $T(y)$. Since T is isotone and coordinate-wise rounding down is isotone, T_η is an order-preserving self-map of the finite grid lattice. It is also succinctly computable from the rational clearing instance.

A fixed point $y = T_\eta(y)$ is an η-approximate fixed point of T: coordinate-wise, $y \leq T(y) < y+\eta$, so $\|T(y)-y\|_\infty \leq \eta \leq \delta$. Transforming back gives a δ-clearing of the original market. Thus bipartite δ-clearing reduces to the succinct Tarski fixed-point problem. Etessami, Papadimitriou, Rubinstein, and Yannakakis show that the succinct Tarski problem lies in both PPAD and PLS [7]. Fearnley, Goldberg, Hollender, and Savani show that CLS = PPAD $\cap$ PLS [8]. Therefore bipartite δ-clearing belongs to CLS.

The same proof covers separable monotone price-impact curves, including independent asset–numeraire constant-product pools; the exact statement and proof are in Appendix A. Routed, cross-asset, or path-dependent AMM execution is outside this extension. For non-affine separable monotone impact, whether this CLS upper bound can be improved to P, or is tight via CLS-hardness, remains open.

Proposition 1 (small-impact polynomial tractability). *In the transformed bipartite system $T(y) = \Pi_{[0,u]}(Ay + c)$, if $\rho(A) < 1$, then the clearing is unique and a δ-clearing can be computed in time polynomial in the input size and $\log(1/\delta)$.*

The proof is the standard weighted-sup-norm contraction argument; Appendix A also gives the Lipschitz version for separable monotone impact.

3.1 Polynomial Time for Subtraction-Free Clearing

The sign-balanced transformation produces the abstract *subtraction-free box fixed-point* problem

$$z = \Pi_{[0,u]}(Bz + g), \qquad B \geq 0, \quad u > 0,$$

where g is the transformed drift. Let z^* denote the least fixed point and write

$$L^* = \{i : z_i^* = 0\}, \qquad I^* = \{i : 0 < z_i^* < u_i\}, \qquad U^* = \{i : z_i^* = u_i\}.$$

The following gives an exact polynomial-time algorithm for affine subtraction-free maps. It does not apply to the non-bipartite hardness instances, where $B \geq 0$ is absent.

We use standard facts about nonnegative matrices [17]: principal-submatrix spectral radius monotonicity, Perron–Frobenius for reducible matrices, Collatz–Wielandt subinvariance, and exact rational M-matrix tests for deciding $\rho(B') < 1$. Appendix A records the associated box-LCP algebra and a counterexample showing why naive subcritical over-capping needs repair.

Lemma 2 (the least fixed point has subcritical interior). *For any $B \geq 0$ and any $g \in \mathbb{R}^n$, the least fixed point of $T(z) = \Pi_{[0,u]}(Bz + g)$ satisfies*

$$\rho(B_{I^*I^*}) < 1.$$

Proof. Suppose $M = B_{I^*I^*}$ has $\rho(M) \geq 1$. Perron–Frobenius, allowing reducibility, gives a nonzero vector $\delta \geq 0$ with $M\delta = \rho(M)\delta$. Extend δ by zero outside I^*. For sufficiently small $\varepsilon > 0$, the point $z(\varepsilon) = z^* - \varepsilon\delta$ remains in the same open box on I^*. On I^*,

$$(Bz(\varepsilon) + g)_{I^*} = z^*_{I^*} - \varepsilon M\delta \leq z^*_{I^*} - \varepsilon\delta = z(\varepsilon)_{I^*}.$$

On U^*, decreasing z^* in nonnegative directions can only decrease the drive, so clipping gives $T_i(z(\varepsilon)) \leq u_i = z_i(\varepsilon)$. On L^*, the fixed-point condition implies $(Bz^* + g)_i \leq 0$, and the drive again only decreases, so $T_i(z(\varepsilon)) = 0 = z_i(\varepsilon)$. Thus $T(z(\varepsilon)) \leq z(\varepsilon)$. The least fixed point of a monotone map is the least prefixed point, so $z^* \leq z(\varepsilon) < z^*$ on the support of δ, a contradiction.

For orientation, the nonnegative-drift case has a one-phase proof: the floor set is determined by reachability, and a single cascade solves the active block. We record this special case in Appendix A.1; it is not used in the proof below. The arbitrary-drift case is harder because floors are no longer reachability-determined, and is handled by a two-phase activation algorithm.

Lemma 3 (floor equals unreachability). *For $g \geq 0$, the positive coordinates of the least fixed point are exactly those reachable from* $\operatorname{supp}(g)$ *in the feeds graph $j \to i$ when $B_{ij} > 0$; the restriction to this reachable set is autonomous and strictly positive.*

Lemma 4 (nonnegative cascade). *An autonomous block with $B \geq 0$, $g \geq 0$, and every coordinate reachable from* $\operatorname{supp}(g)$ *has a unique fixed point, computable in $O(n)$ cascade rounds by exact rational linear solves and standard exact M-matrix tests.*

Theorem 2 (nonnegative-drift SFBFP). *For every rational $B \geq 0$, $g \geq 0$, and $u > 0$, the least fixed point of $z = \Pi_{[0,u]}(Bz + g)$ is computable in time polynomial in the input bit length.*

Proofs of these three special-case statements are in Appendix A.1.

For arbitrary drift, the floor set is not reachability-determined. The device is to split the search into two monotone phases: an outer activation loop and an inner from-above cascade. We use that for a continuous isotone self-map of a box, the Kleene iterates from 0 increase to the least fixed point.

Lemma 5 (from-above cascade for an active block with positive least fixed point). *Let $A \subseteq S$ carry an autonomous block map $T_A(y) = \Pi_{[0,u_A]}(B_{AA}y+g_A)$ with $B_{AA} \geq 0$ and* arbitrary *g_A, and suppose its least fixed point $\zeta = \operatorname{lfp}(T_A)$ is strictly positive. Then ζ is the unique fixed point of T_A and is computed in at most $|A|$ rounds by the from-above cascade that starts with all coordinates clamped ($C = A$) and, in each round with $I = A \setminus C$: solves $\hat{z}_I = (\mathbb{I} - B_{II})^{-1}(g_I + B_{IC}u_C)$; clamps every $i \in I$ with $\hat{z}_i \geq u_i$; otherwise releases from C every c with $(Bz+g)_c < u_c$, halting when none remain.*

Proof. Since $\zeta > 0$, Lemma 2 (applied to T_A) gives $\rho(B_{I_AI_A}) < 1$, where $I_A = \{i \in A : \zeta_i < u_i\}$ and $U_A = \{i \in A : \zeta_i = u_i\}$.

Uniqueness. Any fixed point z' has $\delta = z' - \zeta \geq 0$ (least), and $\delta_i > 0$ forces $\zeta_i < u_i$, so (using $\zeta > 0$) $i \in I_A$; thus $\operatorname{supp}\delta \subseteq I_A$. On $D = \operatorname{supp}\delta$, interiority gives $\delta_D \leq B_{DD}\delta_D$, so Collatz–Wielandt forces $\rho(B_{DD}) \geq 1$, contradicting $\rho(B_{DD}) \leq \rho(B_{I_AI_A}) < 1$ unless $\delta = 0$.

Invariant $C \supseteq U_A$. Initially $C = A \supseteq U_A$. Each repair point satisfies $z \geq \zeta$ (shown below), so a released c (with $(Bz+g)_c < u_c$) has $(B\zeta+g)_c \leq (Bz+g)_c < u_c$, i.e. $\zeta_c < u_c$, hence $c \notin U_A$. Thus $C \supseteq U_A$ throughout and $I = A \setminus C \subseteq I_A$, whence $\rho(B_{II}) < 1$: the interior is always subcritical and $(\mathbb{I} - B_{II})^{-1} \geq 0$.

Positivity by domination. For $C \supseteq U_A$ the true ζ satisfies $\zeta_I = (\mathbb{I}-B_{II})^{-1}(g_I+B_{IC}\zeta_C)$, so $\hat{z}_I - \zeta_I = (\mathbb{I} - B_{II})^{-1}B_{IC}(u_C - \zeta_C) \geq 0$ and $\hat{z}_I \geq \zeta_I > 0$. At a repair round ($\hat{z}_I < u_I$) the point z ($= u$ on C, $= \hat{z}_I \in (0, u_I)$ on I) is therefore prefixed: $T_A(z)_I = \hat{z}_I = z_I$ and $T_A(z)_C \leq u_C = z_C$, so $z \geq \zeta$.

Monotone repair, no re-clamping. From $C = A$ the first round is a repair. If a repair round has a nonempty release set bad, put $C' = C \setminus \text{bad} \supseteq U_A$ and $I' = A \setminus C' \subseteq I_A$; the contraction $T_{C'}(w) = \Pi_{[0,u_{I'}]}(B_{I'I'}w + B_{I'C'}u_{C'} + g_{I'})$ has unique fixed point $\hat{z}_{I'}(C')$, for which the current $z|_{I'}$ is prefixed (equality on the old interior, strict decrease on bad); hence $\hat{z}_{I'}(C') \leq z_{I'}$, so $\hat{z}_c < u_c$ on bad and $\hat{z}_i < u_i$ on the old interior, and the next round is again a repair with $C' \subsetneq C$. After the first repair each round removes at least one coordinate, so within $\leq |A|$ rounds the release set is empty and the returned fixed point equals ζ.

Lemma 6 (monotone activation). *Start from $A = \varnothing$, $z = 0$. Repeat: with $d = Bz + g$, let $\text{act} = \{f \notin A : d_f > 0\}$; if $\text{act} = \varnothing$, stop; otherwise set $A \leftarrow A \cup \text{act}$, recompute $z_A = \operatorname{lfp}(T_A)$ by Lemma 5, and set $z = 0$ off A. This runs for at most $|S|$ iterations, every block it solves has $\operatorname{lfp}(T_A) > 0$, and it returns the least fixed point z^*.*

Proof. Maintain the invariant: $A \subseteq A^* := \operatorname{supp}(z^*)$; $z = \operatorname{lfp}(T_A)$ on A and 0 off A; $\operatorname{lfp}(T_A) > 0$; and $z \leq z^*$. The base case $A = \varnothing$ is vacuous.

Activations are correct. If $d_f = (Bz+g)_f > 0$ with $z \le z^*$, then $(Bz^*+g)_f \ge d_f > 0$, so $z^*_f > 0$, i.e. $f \in A^*$; hence $A \cup \mathrm{act} \subseteq A^*$.

The new block's least fixed point is positive. Write $A' = A \cup \mathrm{act}$. Activation only adds nonnegative input, so the Kleene iterates from 0 for $T_{A'}$ dominate those for T_A on A; hence $\mathrm{lfp}(T_{A'})|_A \ge \mathrm{lfp}(T_A) > 0$. For $f \in \mathrm{act}$, $(B\,\mathrm{lfp}(T_{A'}) + g)_f \ge d_f > 0$, so that coordinate is positive too. Thus $\mathrm{lfp}(T_{A'}) > 0$ and Lemma 5 applies.

$z \le z^*$ is preserved. $z^*_{A'} = \Pi\big(B_{A'A'}z^*_{A'} + B_{A',S\setminus A'}z^*_{S\setminus A'} + g_{A'}\big) \ge \Pi(B_{A'A'}z^*_{A'} + g_{A'}) = T_{A'}(z^*_{A'})$, so $z^*_{A'}$ is prefixed for $T_{A'}$ and $\mathrm{lfp}(T_{A'}) \le z^*_{A'}$.

Termination and correctness. Each productive iteration enlarges A, so there are at most $|S|$. At termination $d_f \le 0$ off A, so $T(z)_f = \Pi(d_f) = 0 = z_f$; on A, z is a fixed point of T_A. Thus $T(z) = z$ and $z \le z^*$; as z^* is least, $z = z^*$.

Theorem 3 (mixed-sign SFBFP is in P). *For every rational $B \ge 0$, every rational drift g of arbitrary sign, and $u > 0$, the least fixed point of $z = \Pi_{[0,u]}(Bz + g)$ is computable in time polynomial in the input bit length.*

Proof. Run the activation loop of Lemma 6: at most $|S|$ activation steps, each invoking one from-above cascade of at most $|S|$ rounds (Lemma 5). Each round is one exact rational linear solve of polynomial bit length together with a standard exact rational M-matrix test for $\rho(B') < 1$ [17]. The total is $O(|S|^2)$ exact solves, polynomial in the input size, and the loop returns z^*.

Corollary 1 (exact polynomial-time clearing for bipartite linear markets). *In the collateral-only linear price-impact model with rational inputs, if the asset borrowing graph G_A is bipartite, then an exact smooth clearing can be computed in polynomial time.*

Proof. Lemma 1 transforms the clearing equations into $y = \Pi_{[0,u]}(Ay + c)$ with $A \ge 0$, preserving rational bit length. Theorem 3 computes the least fixed point of this transformed system in polynomial time. Inverting the sign transform gives an exact clearing of the original linear-impact market.

Remark 1 (the mixed-sign case is settled). Theorem 3 puts the transformed sign-balanced linear map $y = \Pi_{[0,u]}(Ay + c)$, $A \ge 0$, in P for all drift, large impact included. Thus Corollary 1 supersedes the linear-impact CLS upper bound of Theorem 1; the latter remains useful for the separable nonlinear extension of Appendix A. The nonnegative-drift cascade (Theorem 2) is only a special case.

Appendix A records the equivalent upper-bounded Z-matrix complementarity form and the method-level refinements. This also explains why the polynomial algorithms for conservation-like monotone payment networks [1] do not subsume the large-impact DeFi map: the transformed equations are monotone and piecewise-linear, but they are not substochastic payment-allocation equations.

4 Membership and All-or-Nothing Clearing

Proposition 2. *For rational inputs and rational $\delta > 0$, smooth δ-clearing belongs to PPAD.*

Proof. The right-hand sides of the price and liquidation equations are represented by a rational piecewise-linear circuit over addition, multiplication by rational constants, and min/max gates. The price clipping and liquidation clipping make this circuit a continuous self-map of a compact rational box. A δ-clearing is precisely an approximate fixed point of this piecewise-linear map, up to a harmless rescaling of residuals between price and liquidation coordinates. Approximate fixed points of rational piecewise-linear FIXP circuits are in PPAD via the Linear-FIXP/PPAD correspondence [9].

Proposition 3. *The all-or-nothing clearing existence problem belongs to NP. If the signed feedback structure is balanced, all-or-nothing clearing can be found in polynomial time.*

Proof. For NP membership, a certificate is the set of liquidated positions. Given this set, prices are computed directly from (3), and each threshold condition can be checked. In the balanced case, after the sign transform the induced map on $\{0,1\}^m$ is monotone. Starting from the bottom element and iterating the map, the sequence is coordinate-wise nondecreasing in the transformed order. Since every coordinate is binary, each coordinate changes at most once, and the iteration stabilizes after at most m strict changes. This is the finite-lattice analogue of the monotone-clearing algorithms used in classical financial networks; unlike the smooth case, finiteness gives an immediate polynomial bound.

Theorem 4 (all-or-nothing non-bipartite hardness). *The all-or-nothing clearing-existence problem is NP-complete, even for collateral-only linear price-impact markets with rational parameters of polynomial bit complexity, strict threshold margins, bounded liquidation fractions $\alpha_i \leq \varphi C_i$, and connected non-bipartite asset borrowing graph G_A.*

Proof. Membership is Proposition 3. For hardness, reduce from 3-SAT using free bistable variables and a fixed threshold-gate library for NOT, NOR, copy, and dual-rail AND/OR. Appendix B mechanically verifies this finite library with rational parameters, strict margins, close-factor-compatible sales, and the rank-one identity of Lemma 15.

The new part is the sink gadget. From the gate layer compute SAT and $\overline{\mathsf{SAT}}$. Attach a three-position private NOT ring

$$R_1 = 1 - R_2, \qquad R_2 = 1 - R_3, \qquad R_3 = 1 - R_1,$$

and gate R_1 by a controller W driven by $\overline{\mathsf{SAT}}$. The corrected rank-one-realizable controller witness has weights

$$(w_{R_1}, U_W, w_W, u_W) = (1, 4, 4, 1),$$

so $w_{R_1} u_W = U_W$. If the formula is satisfied, then $\overline{\mathsf{SAT}} = 0$, the controller turns on, and the ring has the consistent state verified in Appendix B. If the formula is unsatisfied, then $\overline{\mathsf{SAT}} = 1$, the controller is off, and the sink contains the impossible odd NOT equation $r = \mathrm{NOT}(r)$. Hence clearings are in bijection with satisfying assignments.

The ring positions use assets a_1, a_2, a_3 and contribute the triangle $a_1 - a_2 - a_3 - a_1$ to G_A; the controller edge $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$ attaches it to the SAT component. Thus the hard instances have connected non-bipartite asset graph.

5 PPAD-Hardness in the Non-Bipartite Regime

Our hardness proof uses only two generalized-circuit gates. Let $T_{[0,1]}$ denote clipping to $[0, 1]$. A κ-approximate assignment gives each wire a value in $[0, 1]$ and makes every gate output differ from its prescribed value by at most κ. This is a local notion of satisfaction: circuits may contain cycles, and no topological evaluation order is assumed.

Lemma 7 (SUM/NOT source). *There is a constant $\kappa > 0$ such that finding a κ-approximate solution to generalized circuits with only the gates*

$$G_+(u, w) : y_v = T_{[0,1]}(y_u + y_w), \qquad G_{1-}(u) : y_v = 1 - y_u$$

is PPAD-complete, even with fan-out two.

We use Budish et al. [6, Appendix D, Definition 10 and Remark 7] as the load-bearing source; related restricted-gate hardness appears in Filos-Ratsikas et al. [5]. Since truncated-SUM is already primitive, our gadgets need no high-gain chains, comparison gates, or band resets.

Lemma 8 (unbalanced SUM/NOT source). *Lemma 7 remains PPAD-hard under the promise that the signed dependency graph of the source circuit contains a cycle with an odd number of NOT gates in the same connected component as an original hard-instance wire.*

Proof. Given a SUM/NOT circuit C, pick any wire s in it. Add two fresh wires a, b and the two gates

$$a = 1 - s, \qquad b = T_{[0,1]}(s + a).$$

Every solution of the augmented circuit restricts to a solution of C, because the original gates are unchanged. Conversely, any exact solution of C extends by setting $a = 1 - s$ and $b = 1$. The new dependency graph contains the cycle $s - a - b - s$, with the edge $s - a$ coming from a NOT gate and the two edges incident to b coming from the SUM gate. Thus the cycle has odd NOT parity and lies in the connected component containing s. A κ-approximate solution of the augmented circuit gives a κ-approximate solution of C by projection, so a solver for this promised subclass would solve the PPAD-hard source problem.

Wire encoding and writer equation. For each circuit wire v, create an asset and encode $y_v = 2 - p_v$, with intended prices in $[1, 2] \subset [0, 3]$. A collateral-only writer with collateral output O, debt input U, and parameters C, D, ε satisfies

$$x = \Pi_{[0,1]}(c - a y_U + \lambda y_O), \qquad a = D/\varepsilon, \quad \lambda = C/\varepsilon, \quad c = (2D - 2C)/\varepsilon. \quad (7)$$

If this writer is the unique effective writer of O, then $y_O = x$. All gadgets have self coefficient at most $1/2$, so each local fixed point is unique and residuals are amplified only by a universal constant. Close-factor compliance is mechanical: desired output weight η is implemented by taking $\alpha = \varphi/2$ and output slope $s_O = 2\eta/\varphi$; Appendix D records the fixed parameter sheet.

The two gadgets. The NOT gate $y_O = 1 - y_U$ uses one writer with $C = 1, D = 2, \varepsilon = 3$, giving

$$y_O = \Pi_{[0,1]}\left(\frac{2 - 2y_U + y_O}{3}\right),$$

whose unique fixed point is $1 - y_U$. The truncated-SUM gate is a constant-size macro. First, two co-writers, each contributing half a unit of price drop and reading the complement wires $\bar{U}, \bar{W}$, create an average asset A satisfying $y_A = (y_U + y_W)/2$. Second, a NOT gadget creates $\bar{A}$. Finally, one writer with $C = 1, D = 2, \varepsilon = 2$ reads $\bar{A}$ and satisfies

$$y_O = \Pi_{[0,1]}\left(y_A + \frac{1}{2} y_O\right),$$

whose unique fixed point is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. The algebra and constant error bounds are recorded in Appendix D. In particular, a source gate violation is at most $B\delta$ for an absolute constant B independent of the circuit size.

Lemma 9 (forward decoding). *For every SUM/NOT circuit C, one can construct in polynomial time a collateral-only liquidation instance I_C such that every δ-clearing of I_C decodes via $y_v = 2 - p_v$ to a $B\delta$-satisfying assignment of C, where $B = O(1)$ is independent of $|C|$.*

Proof. Replace each source gate by the corresponding constant-size macro. Debt-leg fan-out is non-invasive because only collateral sales enter (3). Every public wire has a unique effective writer, except for the two intended co-writers in the average macro. Since each macro has constant size and self coefficient bounded away from one, the local decoding error is at most $B\delta$ for a universal B. Generalized-circuit satisfaction is local, so cycles require no topological induction.

The target search problem is total by Brouwer, because clipping makes the clearing map continuous on a compact box. A total-search reduction therefore needs only forward decoding.

Lemma 10 (gadget parity). *Let H_C be the signed undirected dependency graph of a SUM/NOT circuit C, with a negative edge for every NOT gate and positive edges from each input of a SUM gate to its output. For the liquidation instance I_C constructed above, with no additional disconnected anchor, the asset borrowing graph $G_A(I_C)$ is bipartite if and only if H_C is sign-balanced. In particular, an odd-NOT cycle in C maps to an odd cycle in $G_A(I_C)$.*

Proof. The claim is a parity accounting for the templates. A NOT gate is one collateral-debt edge, hence odd. A SUM input incidence is the even path

$$U - \bar{U} - A - \bar{A} - O,$$

and the same holds for $W \to O$; the remaining internal SUM subgraph is bipartite. Hence $G_A(I_C)$ is obtained from H_C by replacing negative edges by odd paths and positive edges by even paths, plus bipartite subdivisions. Cycle parity is therefore preserved modulo the number of NOT edges. By Harary's criterion, $G_A(I_C)$ is bipartite exactly when every source cycle has even NOT parity. The odd-NOT cycle from Lemma 8 is thus an intrinsic non-bipartite cycle in the hard liquidation instance.

Theorem 5. *There exists a constant $\delta_0 > 0$ such that, in the non-bipartite regime, finding a δ_0-approximate clearing in collateral-only linear price-impact liquidation markets is PPAD-hard. Consequently, for a sufficiently small fixed δ_0, the problem is PPAD-complete.*

Proof. Let κ be the constant from Lemma 8 and let B be the constant from Lemma 9. Choose $\delta_0 = \kappa/(10B)$. Given a promised unbalanced SUM/NOT circuit C, build I_C with no disconnected anchor. By Lemma 10, the asset borrowing graph of I_C is non-bipartite. Any δ_0-clearing decodes to a κ-satisfying assignment of C. Thus a clearing oracle for non-bipartite instances would solve a PPAD-complete source problem. Membership follows from the PPAD proposition.

The quantifiers in Theorem 5 are worst-case. We do not claim that every non-bipartite borrowing graph makes clearing hard. Rather, Lemma 10 shows that the constructed hard instances are non-bipartite for an intrinsic reason: the source circuit contains odd-NOT frustration, and the liquidation gadgets preserve that parity as an odd collateral-debt cycle. The non-bipartiteness is therefore part of the hard component, not a disconnected certificate attached after the fact.

Related work. Prior financial clearing already exposes Tarski/lattice structure [14, 15]; fire-sale games show related lattice phenomena [2]. Besting, Hoefer, and Huth [1] go further for generalized Eisenberg–Noe payment networks: they compute extremal Tarski fixed points in polynomial time, and the minimal one strongly polynomially. That setting is payment allocation with conservation structure in the Eisenberg–Noe core. Our equations are price-impact equations:

collateral sales move prices, assets may serve as both collateral and debt, and the large-impact transformed matrix can have spectral radius at least one. The new point is the sign boundary, not generic "least fixed point in P": bipartite feedback becomes subtraction-free (P for linear impact, CLS for separable nonlinear impact), while odd collateral-debt cycles implement NOT and give PPAD-hardness.

CDS hardness uses derivative contingencies [16, 4]; here NOT comes from collateral-only debt relief. DeFi liquidation and lending work supplies the economic setting [10–13, 3]. Appendix G records method-level refinements around the exact SFBFP algorithm; it is not needed for the main P/CLS/PPAD boundary.

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A Additional Details for Sign-Balanced Clearing

This appendix records mechanical extensions and algebraic details used around the sign-balanced side. None of the main support theorems depends on the exploratory material in Appendix G.

Proof (Proof of Proposition 1). By Perron–Frobenius, since $A \geq 0$ and $\rho(A) < 1$, there are $v > 0$ and $\lambda < 1$ with $Av \leq \lambda v$. In the weighted sup norm $\|r\|_v = \max_i |r_i|/v_i$, the projection $\Pi_{[0,u]}$ is nonexpansive coordinate-wise and

$$\|T(y) - T(y')\|_v \leq \|A(y - y')\|_v \leq \lambda \|y - y'\|_v.$$

Thus T is a contraction, so Banach iteration converges geometrically to the unique fixed point and gives a δ-clearing in time polynomial in the input size and $\log(1/\delta)$.

A.1 Nonnegative-Drift Special Case

Proof (Proof of Lemma 3). Let A be the set of coordinates reachable from $\mathrm{supp}(g)$ in the feeds graph, and L its complement. In the Kleene iteration from 0, a coordinate becomes positive exactly when it receives positive input from g or from an already positive coordinate, since all coefficients and all existing positive values are nonnegative. Hence the eventual positive coordinates are exactly A. If a coordinate in A fed a coordinate in L, then the latter would also be reachable; therefore the restricted active block on A is autonomous.

Proof (Proof of Lemma 4). Write S for the block. By Lemma 3, $z^* > 0$, so $I^* = \{i : z_i^* < u_i\} = S \setminus U^*$; by Lemma 2, $\rho(B_{I^*I^*}) < 1$. Maintain a tentative clamp set C, initially empty, put $I = S \setminus C$, and when $\rho(B_{II}) < 1$ write

$$\hat{z}_I(C) = (I - B_{II})^{-1}(g_I + B_{IC}u_C).$$

One round executes the first applicable case:

(G1) if $\rho(B_{II}) \geq 1$: move any one coordinate of I into C;
(G2) else if $\hat{z}_i(C) \geq u_i$ for some $i \in I$: move all such i into C;
(R) else form z with $z_C = u_C$, $z_I = \hat{z}_I(C)$, and release bad $= \{c \in C : (Bz + g)_c < u_c\}$, halting if this set is empty.

Growth terminates because enlarging C only shrinks the principal submatrix B_{II}. Every repair point is prefixed: $z_I = \hat{z}_I(C) \geq 0$ by the Neumann series, $(Bz+g)_I = z_I$, and $T(z)_C \leq u_C = z_C$; hence $z \geq z^*$. Thus a released coordinate cannot be in the true clamp set U^*. After the first repair we have $C \supseteq U^*$, hence the next interior $I \subseteq I^*$ is subcritical. The old repair point is prefixed for the newly released subproblem, so the next solve is coordinate-wise no larger; released coordinates stay below u, no re-clamping occurs, and C strictly shrinks on each later repair. Hence the cascade halts after at most $2|S|$ rounds.

At halt $T(z) = z$. If $z' \geq z^*$ is another fixed point and $D = \{i : z'_i > z^*_i\}$, then $D \subseteq I^*$ and $\delta_D \leq B_{DD}\delta_D$ for $\delta = z' - z^* > 0$ on D. Collatz–Wielandt gives $\rho(B_{DD}) \geq 1$, contradicting $D \subseteq I^*$ and $\rho(B_{I^*I^*}) < 1$. Thus the halted fixed point is unique and equals z^*.

Proof (Proof of Theorem 2). Compute the reachable set A of Lemma 3 by graph search and set $z^*_L = 0$. The active subsystem on A is autonomous and satisfies the hypotheses of Lemma 4, so the cascade returns its unique fixed point in $O(|A|)$ rounds. The exact linear solves have polynomial bit complexity, and each subcriticality decision is a standard exact rational M-matrix test. Combining the active solution with $z^*_L = 0$ gives the least fixed point of the original system.

Proposition 4 (when the transformed drift is nonnegative). *Suppose G_A is bipartite. The transformed drift c of Lemma 1 can be made nonnegative exactly in the two-sector case in which no asset is used both as collateral and as debt, the bipartition is chosen with all collateral assets in L and all debt assets in R, and*

$$p^0_k \geq s_k \sum_{i:a_i=k} \alpha_i \quad (k \in L), \qquad p^0_k \leq M_k \quad (k \in R), \qquad \varepsilon_i \geq \theta_i D_i M_{b_i} \quad (i \in [m]).$$

In that regime the clearing is computable in polynomial time by Theorem 2, even when the price impact is large ($\rho(A) \geq 1$).

Proof (Proof of Proposition 4). With all collateral assets in L and all debt assets in R, the four cases of Lemma 1 give

$$c_k = \begin{cases} p^0_k - s_k \sum_{i:a_i=k} \alpha_i, & k \in L, \\ M_k - p^0_k, & k \in R, \end{cases} \qquad c_i = 1 - \frac{\theta_i D_i M_{b_i}}{\varepsilon_i}.$$

The displayed inequalities are exactly nonnegativity of these entries. If an asset is used both as collateral and as debt, then under any bipartition one witnessing position has $a_i \in R$, and Lemma 1 gives the strictly negative drift $-C_i M_{a_i}/\varepsilon_i$. Hence no bipartition can make $c \geq 0$.

Remark 2. The two-sector condition is strictly stronger than bipartiteness: it forbids the asset role-reversal that creates the debt-relief feedback of Fig. 1. Theorem 2 is therefore a useful special case of the general algorithm, not the boundary result.

Gauge normal form for balanced SUM/NOT.

Lemma 11 (gauge normal form). *Let $y_v = \Pi_{[0,1]}(y_u + y_w)$ be a truncated-SUM gate and write $\bar{t} = 1 - t$. Then*

$$\bar{y}_v = 1 - \Pi_{[0,1]}(y_u + y_w) = \max\{1 - y_u - y_w, 0\} = \Pi_{[0,1]}(\bar{y}_u + \bar{y}_w - 1).$$

Consequently, applying $y \mapsto 1 - y$ to a Harary-balanced subset of wires turns a SUM/NOT circuit into a subtraction-free clipped-affine system $y = \Pi_{[0,1]}(Ay + c)$ with $A \geq 0$.

Proof. The identity uses $1-\Pi_{[0,1]}(t) = \Pi_{[0,1]}(1-t)$ on the relevant range. A balanced signed dependency graph admits a vertex flip making every edge positive. Performing $y \mapsto 1-y$ on the flipped wires replaces each NOT by an identity and each SUM by a gate of the displayed form, whose variable coefficients are nonnegative.

Corollary 2 (balanced circuit fragment). *The sign-balanced SUM/NOT fragment is the circuit analogue of the transformed DeFi system in Lemma 1: after a gauge transform it has the form $y = \Pi(Ay+c)$ with $A \geq 0$. Therefore the balanced fragment is polynomial-time solvable by Theorem 3; SUM/NOT hardness must come from sign-imbalance.*

Separable monotone impact and comparative statics.

Theorem 6 (separable monotone price impact). *Replace the linear price equation by*

$$p_k = g_k\left(\sum_{i:a_i=k} \alpha_i x_i\right),$$

where each $g_k : [0,\infty) \to [0, M_k]$ is continuous, nonincreasing, and polynomial-time evaluable on rational inputs to any requested precision. If G_A is bipartite, then for every rational $\delta > 0$, smooth δ-clearing reduces in polynomial time to succinct Tarski and hence belongs to CLS. This includes independent asset–numeraire constant-product pools such as $g_k(Q) = \min\{\kappa_k/(R_k+Q), M_k\}$.

Proof (Proof of Theorem 6). Use the same bipartition (L, R) and transformed variables q, z as in Theorem 1. For the price block, if $k \in L$, then $x_i = 1 - z_i$ for positions collateralized by k, so the sale quantity is nonincreasing in each z_i; since g_k is nonincreasing, $q_k = p_k$ is nondecreasing. If $k \in R$, then $x_i = z_i$, the sale quantity is nondecreasing, $p_k = g_k(\cdot)$ is nonincreasing, and $q_k = M_k - p_k$ is nondecreasing. The liquidation block is the same four-case computation as Lemma 1, so the transformed map is isotone. The grid-rounding proof of Theorem 1 applies.

Proposition 5 (sign-balanced comparative statics). *Fix a grid used in the proof of Theorem 1. Suppose two sign-balanced instances have transformed grid maps T_η and S_η with $T_\eta(y) \leq S_\eta(y)$ for every grid point y. Then the least grid clearing of T_η is no larger than the least grid clearing of S_η, and the same monotone order holds for the greatest grid clearings.*

Proof. On a finite lattice, the least fixed point of an isotone map is obtained by iterating from the bottom element, and the greatest fixed point by iterating from the top. The pointwise inequality and isotonicity imply the comparison by induction.

Proposition 6 (Lipschitz tractability). *In the separable monotone-impact model of Theorem 6, suppose each g_k is γ_k-Lipschitz on the relevant sales range.*

Form the nonnegative gain matrix

$$\widehat{A} = \begin{pmatrix} 0 & \widehat{P} \\ \widehat{Q} & 0 \end{pmatrix}, \qquad \widehat{P}_{ki} = \gamma_k \alpha_i \ (a_i = k), \quad \widehat{Q}_{ia_i} = \frac{C_i}{\varepsilon_i}, \quad \widehat{Q}_{ib_i} = \frac{\theta_i D_i}{\varepsilon_i}.$$

If $\rho(\widehat{A}) < 1$, then the transformed clearing map is a contraction in a weighted sup norm, the clearing is unique, and a δ-clearing is computable in time polynomial in the input size and $\log(1/\delta)$.

Proof. The clipping maps are coordinate-wise nonexpansive, and the transformed price and liquidation blocks are Lipschitz-bounded by $\widehat{P}$ and $\widehat{Q}$, so $|T(y) - T(y')| \le \widehat{A}|y - y'|$. Perron–Frobenius gives a weighted sup norm in which the Lipschitz factor is below one.

Box-LCP and exact subcriticality tests.

Lemma 12 (algebraic box-LCP form). *For $B \ge 0$ and any g, the least fixed point z^*, its floor set L^*, and its clamp set U^* are described by a finite system of linear equations and rational inequalities:*

$$z^*_{L^*} = 0, \qquad z^*_{U^*} = u_{U^*}, \qquad z^*_{I^*} = (I - B_{I^*I^*})^{-1}\big(g_{I^*} + B_{I^*U^*}u_{U^*}\big),$$

together with the lower-, interior-, and upper-bound inequalities defining the three sets.

Proof. On I^*, the clipping is inactive, and the inverse exists by Lemma 2. The boundary inequalities are exactly the box complementarity conditions.

Lemma 13 (exact subcriticality tests). *For rational $B' \ge 0$, the condition $\rho(B') < 1$ can be tested by exact rational polynomial-time M-matrix tests for $I - B'$. No numerical eigenvalue approximation or transcendence bound is needed.*

Proof. $I - B'$ is a Z-matrix, and $\rho(B') < 1$ iff $I - B'$ is a nonsingular M-matrix iff there is $v > 0$ with $(I - B')v > 0$. After normalization, this is a rational linear feasibility problem; equivalent principal-minor or elimination tests give the same exact decision.

Lemma 14 (subcritical upper bound). *Assume $\rho(B_{SS}) < 1$ and $g' \ge 0$. The unclipped solve $\hat{z} = (I - B_{SS})^{-1}g'$ dominates the least box fixed point on S. If $\hat{z} \le u_S$, then $\hat{z}$ is exact on S. Otherwise, the over-threshold set can strictly contain the true clamp set.*

Proof. The clipped Kleene orbit from 0 is bounded above by the unclipped affine orbit, which increases to $\hat{z}$. Strict over-capping occurs for

$$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad g' = (0, 10), \qquad u = (5, 1),$$

where $\hat{z} = (10, 10)$ but the box fixed point is $(1, 1)$.

For $y = \Pi_{[0,u]}(Ay + c)$, define $B = I - A$ and $d = -c$. Then $F(y) = By + d$. Coordinate-wise, the fixed-point condition is equivalent to

$$y_i = 0 \Rightarrow F_i(y) \geq 0, \quad 0 < y_i < u_i \Rightarrow F_i(y) = 0, \quad y_i = u_i \Rightarrow F_i(y) \leq 0.$$

Introducing an upper-bound multiplier $w \geq 0$ gives the upper-bounded LCP formulation

$$0 \leq y \perp By + d + w \geq 0, \qquad 0 \leq w \perp u - y \geq 0.$$

Indeed, if $0 < y_i < u_i$, then $w_i = 0$ and $(By + d)_i = 0$. If $y_i = 0$ and $u_i > 0$, then $w_i = 0$ and $(By + d)_i \geq 0$. If $y_i = u_i$, then $(By + d)_i = -w_i \leq 0$. Coordinates with $u_i = 0$ may be eliminated in preprocessing.

The ordinary standard-LCP embedding of this upper-bounded system, with variables (y, w), has matrix

$$\begin{pmatrix} B & I \\ -I & 0 \end{pmatrix}.$$

The $+I$ block is forced by the sign of the upper-bound multiplier and gives positive off-diagonal entries. Hence the bounded system should not be described as an ordinary Z-matrix LCP. What bipartiteness gives unconditionally is the monotone clipped-affine form, the Tarski reduction, and the upper-bounded Z-matrix base structure. A polynomial-time algorithm follows in the small-impact regime where $\rho(A) < 1$, equivalently $I - A$ is a nonsingular M-matrix.

B All-or-Nothing NP-Hardness Construction

This appendix gives the construction used in Theorem 4. The order matters: first we record the realizability constraints of the liquidation model, then the fixed gate library, and only then the SAT-gated odd-NOT ring. In particular, we never claim that arbitrary signed threshold matrices are implementable.

B.1 Discrete Clearing as a Signed Threshold Network

Fix a collateral-only linear price-impact instance with positions $i \in [m]$, assets k, parameters $a_i, b_i, C_i, D_i, \theta_i, \alpha_i, p_k^0, s_k, M_k$, with $a_i \neq b_i$ and $0 \leq \alpha_i \leq \varphi C_i$. In the all-or-nothing model $x_i \in \{0, 1\}$ and

$$p_k = \Pi_{[0,M_k]}\left(p_k^0 - s_k \sum_{j:a_j=k} \alpha_j x_j\right).$$

When clipping does not bind, a clearing is a fixed point of the signed threshold update

$$x_i = \mathbf{1}[h_i(x) > 0], \qquad h_i(x) = \beta_i + \sum_j W_{ij} x_j, \tag{8}$$

where

$$\beta_i = \theta_i D_i p^0_{b_i} - C_i p^0_{a_i}, \qquad W_{ij} = \begin{cases} C_i s_{a_i} \alpha_j, & a_j = a_i, \\ -\theta_i D_i s_{b_i} \alpha_j, & a_j = b_i, \\ 0, & \text{otherwise.} \end{cases}$$

The positive terms are shared-collateral effects, and the negative terms are debt-relief reads. The self-coupling $W_{ii} = C_i s_{a_i} \alpha_i$ is positive.

Lemma 15 (rank-one positive blocks). *If positions i and j share a collateral asset a, then*

$$\begin{pmatrix} W_{ii} & W_{ij} \\ W_{ji} & W_{jj} \end{pmatrix} = \begin{pmatrix} C_i s_a \alpha_i & C_i s_a \alpha_j \\ C_j s_a \alpha_i & C_j s_a \alpha_j \end{pmatrix}$$

has rank one. In particular $W_{ii} W_{jj} = W_{ij} W_{ji}$. For the controller-ring pair below, this identity is

$$U_W u_W = w_{R_1} w_W.$$

Debt-read coefficients are not constrained by this identity.

Proof. The two columns are proportional with ratio α_j/α_i, so the determinant is zero. Debt reads go through the debt asset and can be scaled by $\theta_i D_i$ and the corresponding sale impact of the read asset, rather than through the shared collateral price.

Lemma 16 (band/no clipping). *For each asset k, let $\Sigma_k = \sum_{j:a_j=k} \alpha_j$. Set*

$$p^0_k = 2 s_k \Sigma_k + 1, \qquad M_k = p^0_k + 1.$$

Then for every $x \in \{0,1\}^m$, the price $p_k = p^0_k - s_k \sum_{j:a_j=k} \alpha_j x_j$ lies strictly inside $[0, M_k]$, so the outer clipping in (3) never binds. If the gadget biases are placed with margins at least one, the tie convention at $h_i = 0$ is irrelevant.

Proof. The price lies in $[p^0_k - s_k \Sigma_k, p^0_k] = [s_k \Sigma_k + 1, 2 s_k \Sigma_k + 1] \subset (0, M_k)$. The threshold margins are then checked in the affine system (8).

B.2 A Fixed DeFi-Realizable Gate Library

The construction uses dual rails: a signal u has an asset for u and a separate asset for $\bar{u}$. Free bits are bistable positions, and NOT gates create complementary rails.

Lemma 17 (fixed threshold gate library). *The following constant-size gates are realizable in the all-or-nothing DeFi model with fresh bus and output assets, polynomial-bit rational parameters, strict margins, and $\alpha_i \leq \varphi C_i$: a free bit, NOT, copy/literal, fan-in-k NOR via a fresh bus, the resulting OR/AND gates by dual rail, the SAT bit, and the gated ring of Lemma 20. Shared-collateral uses obey Lemma 15.*

Proof. A free bit P_v uses a unique collateral asset and an inert debt anchor, with $h_v = \beta_v + w_{vv}x_v$ and $-w_{vv} < \beta_v < 0$; both $x_v = 0$ and $x_v = 1$ are consistent. A NOT gate $Q = \mathrm{NOT}(P_v)$ uses debt asset σ_v and fresh collateral, with

$$h_Q = \beta_Q + w_Q x_Q - W_Q x_v,$$

and $0 < \beta_Q < W_Q - w_Q$, so the input swing dominates self-feedback.

For a NOR gate, route all inputs onto a fresh bus asset β_g. A copy-writer for input t sells the bus exactly when the corresponding input is true; the gate position reads the bus as debt and has a fresh private collateral asset. Its threshold is $h_g = \beta_g + w_g x_g - \sum_t c_t u_t$. Choose the bias in a gap of the attainable sums $\sum_t c_t u_t$ and choose w_g below the gap. The gate is then uniquely consistent and equals the intended downward threshold; OR and AND follow by dual rails and NOTs. Copy-writers sharing the bus obey the rank-one identity, while their faithfulness is maintained by scaling the relevant debt-read swings. The SAT bit is an acyclic composition of these fixed gates. All parameters use a fixed finite set of rational weights, except for polynomially many fresh copies, and the band choice in Lemma 16 prevents clipping.

Remark 3. The lemma is deliberately a fixed-library statement, not an arbitrary signed-matrix realizability theorem. This is the point of the rank-one discipline: only the gates actually used by the reduction are implemented.

B.3 The SAT-Gated Odd-NOT Ring

Build an odd ring with three positions

$$R_2 = (a_2; a_1), \qquad R_3 = (a_3; a_2), \qquad R_1 = (a_1; a_3),$$

so the ungated ring enforces $R_2 = \mathrm{NOT}(R_1)$, $R_3 = \mathrm{NOT}(R_2)$, and $R_1 = \mathrm{NOT}(R_3)$, i.e. $R_1 = \mathrm{NOT}(R_1)$. This has no consistent state and gives the odd asset cycle $a_1 - a_2 - a_3 - a_1$. A controller W shares collateral a_1 with R_1 and reads $\sigma_{\overline{\mathsf{SAT}}}$ as debt:

$$h_W = \beta_W + w_W x_W + u_W x_{R_1} - V_W x_{\overline{\mathsf{SAT}}},$$

$$h_{R_1} = \beta_{R_1} + w_{R_1} x_{R_1} - V_{R_1} x_{R_3} + U_W x_W.$$

The shared-collateral terms obey $U_W u_W = w_{R_1} w_W$. We require

$$\begin{aligned} \mathsf{SAT} = 1: &\quad \beta_W > 0 \Rightarrow W = 1,\\ \mathsf{SAT} = 0: &\quad V_W > \beta_W + w_W + u_W \Rightarrow W = 0,\\ W = 0: &\quad 0 < \beta_{R_1} < V_{R_1} - w_{R_1} \quad \text{(clean NOT)},\\ W = 1: &\quad U_W > V_{R_1} - \beta_{R_1} \Rightarrow R_1 = 1. \end{aligned}$$

Lemma 18 (rank-one-correct witness). *The domination inequalities and rank-one identity are simultaneously satisfied by*

$$(\beta_{R_1}, w_{R_1}, V_{R_1}, U_W) = (1, 1, 3, 4), \qquad (\beta_W, w_W, u_W, V_W) = (1, 4, 1, 7).$$

Equivalently, the shared-collateral tuple is $(w_{R_1}, U_W, w_W, u_W) = (1, 4, 4, 1)$.

Proof. Rank one holds because $U_W u_W = 4 \cdot 1 = 4 = 1 \cdot 4 = w_{R_1} w_W$. The ring inequalities give $0 < 1 < 3 - 1 = 2$ and $4 > 3 - 1 = 2$. The controller inequalities give $1 > 0$ and $7 > 1 + 4 + 1 = 6$.

Lemma 19 (close-factor realization). *The weights in Lemma 18 are realized with $C_{R_1} = C_W = 1$,*

$$\alpha_{R_1} = \frac{\varphi}{8}, \qquad \alpha_W = \frac{\varphi}{2}, \qquad s_{a_1} = \frac{8}{\varphi}.$$

Then $w_{R_1} = 1$, $U_W = 4$, $u_W = 1$, and $w_W = 4$, while $\alpha_{R_1}, \alpha_W \leq \varphi C$. The debt reads $V_{R_1} = 3$ and $V_W = 7$ are realized independently through a_3 and $\sigma_{\overline{\mathsf{SAT}}}$, by scaling θD and the corresponding slopes. The biases are set by the p^0's, and Lemma 16 keeps all prices in band.

Lemma 20 (ring behavior). *Under Lemmas 18 and 19, the block $\{W, R_1, R_2, R_3\}$ has the unique consistent state*

$$(x_W, x_{R_1}, x_{R_2}, x_{R_3}) = (1, 1, 0, 1)$$

when $\mathsf{SAT} = 1$. It has no consistent state when $\mathsf{SAT} = 0$.

Proof. If $\mathsf{SAT} = 1$, then $\overline{\mathsf{SAT}} = 0$, so the controller inequality forces $x_W = 1$. The $W = 1$ inequality forces $x_{R_1} = 1$, and the clean NOTs force $x_{R_2} = 0$ and $x_{R_3} = 1$; rechecking W and R_1 gives consistency and uniqueness. If $\mathsf{SAT} = 0$, then $\overline{\mathsf{SAT}} = 1$, so $x_W = 0$. The ring then reduces to three clean NOTs and hence to $R_1 = \mathrm{NOT}(R_1)$, impossible.

B.4 Reduction from SAT

Given a 3-CNF formula ϕ over variables $v_1, \ldots, v_n$, build free bits $P_{v_1}, \ldots, P_{v_n}$, a NOT per variable for dual rails, an acyclic gate circuit computing

$$\mathsf{SAT} = \bigwedge_c (\ell_{c,1} \vee \ell_{c,2} \vee \ell_{c,3})$$

and $\overline{\mathsf{SAT}}$, and the gated ring above driven by $\overline{\mathsf{SAT}}$. All parameters are chosen according to Lemmas 16, 17, 18, and 19.

Theorem 7 (correctness of the discrete construction). *The constructed instance I_ϕ has an all-or-nothing clearing iff ϕ is satisfiable. Moreover, the number of clearings equals the number of satisfying assignments.*

Proof. By Lemma 16, all updates are affine threshold updates. Free bits are mutually independent and bistable. The literal, clause, SAT, and $\overline{\mathsf{SAT}}$ layers form an acyclic functional circuit over the free bits by Lemma 17. The gated ring is a sink: it is read by no upstream gate. Thus a global fixed point is exactly a choice of free-bit assignment, its unique functional completion, and a consistent ring state. By Lemma 20, the ring has a consistent state exactly when $\mathsf{SAT} = 1$, and then that state is unique. This gives the bijection between clearings and satisfying assignments.

Lemma 21 (connected non-bipartite asset graph). *The asset borrowing graph $G_A(I_\phi)$ contains the odd triangle $a_1 - a_2 - a_3 - a_1$, and the controller edge $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$ attaches this triangle to the component carrying the SAT circuit and variables. Hence the relevant component is connected and non-bipartite.*

Proof. The three NOTs in the ring give the edges $\{a_1, a_2\}$, $\{a_2, a_3\}$, and $\{a_3, a_1\}$. The controller $W = (a_1; \sigma_{\overline{\mathsf{SAT}}})$ adds $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$. The SAT circuit is connected to $\sigma_{\overline{\mathsf{SAT}}}$ by construction. Adding edges to a graph containing an odd cycle preserves non-bipartiteness.

Theorem 8 (Theorem 4, restated). *All-or-nothing clearing existence is NP-complete under the restrictions stated in Theorem 4. If the signed coupling graph is balanced, a clearing can be found in polynomial time as in Proposition 3.*

Proof. Membership is Proposition 3. For hardness, the construction has size polynomial in $|\phi|$, rational coefficients of polynomial bit complexity, strict margins, and $\alpha_i \leq \varphi C_i$. Theorem 7 shows that I_ϕ clears iff ϕ is satisfiable, and Lemma 21 shows that the hard instances are connected and non-bipartite. This is a polynomial many-one reduction from 3-SAT.

Remark 4 (computational sanity check). We additionally validated the gadget truth tables and the rank-one realizability constraint with brute-force fixed-point enumerators on small instances. The proof above rests on the inequalities; the checkers are only a sanity-check artifact.

C Discrete Parity Ledger

This optional structural appendix mirrors the smooth SUM/NOT parity lemma. It is not needed for Theorem 4; its role is to show that in the discrete construction, imbalance is localized to the gated odd-NOT ring.

Definition 1 (signed coupling graph). *This is the undirected symmetrization of the diagnostic position graph from Section 2. The vertices of G_X are positions. For $i \neq j$, add a positive edge when $a_i = a_j$ and a negative edge when $a_i = b_j$ or $a_j = b_i$. Sharing only a debt asset induces no edge.*

Lemma 22 (bipartite implies balanced). *If G_A is bipartite, then G_X is balanced.*

Proof. Let χ be a two-coloring of G_A, and set $\sigma_i = (-1)^{\chi(a_i)}$. If $a_i = a_j$, then $\sigma_i\sigma_j = +1$. If $a_j = b_i$ or $a_i = b_j$, then the collateral and debt assets of the corresponding position have opposite colors, so $\sigma_i\sigma_j = -1$. Thus edge signs equal $\sigma_i\sigma_j$.

Lemma 23 (converse under unique effective writers). *Suppose every priced asset is the collateral of exactly one effective writer. Suppose also that all other debt anchors are private pendant assets. If G_X is balanced, then G_A is bipartite.*

Proof. Let σ_i be a balancing gauge. For each priced asset k, color it by the sign of its unique writer: $(-1)^{\chi(k)} = \sigma_{w(k)}$. For a position i, its collateral a_i has writer i, so $(-1)^{\chi(a_i)} = \sigma_i$. If b_i is priced with writer j, then $a_j = b_i$, so $\{i, j\}$ is a negative edge and balance gives $\sigma_j = -\sigma_i$; hence a_i and b_i receive opposite colors. If b_i is an inert private anchor, color it opposite a_i. Thus every edge of G_A crosses the cut.

Theorem 9 (localized frustration). *In I_ϕ, deleting the three ring positions R_1, R_2, R_3 leaves a balanced signed coupling graph and a bipartite asset graph. The imbalance is localized to the gated-ring block: R_1, R_2, R_3 form an odd-NOT triangle, while the controller W has the side-couplings needed to gate R_1 and attaches that block to the SAT circuit. In $G_A(I_\phi)$, W contributes only the asset edge $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$; the triangle $a_1 - a_2 - a_3 - a_1$ is the unique odd cycle of the ring block, and $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$ connects it to the feed-forward component.*

Proof. The feed-forward gate library is acyclic in the Boolean sense and can be assigned the layer-parity gauge: cross-layer debt-read edges are negative and same-bus shared-collateral edges are positive. Lemma 22 gives the matching bipartite coloring of its asset graph. In G_X, the ring contributes the three negative edges R_1R_2, R_2R_3, R_3R_1, producing a negative 3-cycle. The controller contributes a positive shared-collateral edge WR_1, the negative side-coupling WR_2 because W's collateral asset a_1 is R_2's debt asset, and a negative debt-read attachment to the writer of $\sigma_{\overline{\mathsf{SAT}}}$ in the SAT component. Removing R_1, R_2, R_3 removes the ring obstruction and the WR_1, WR_2 side-couplings; the remaining controller attachment is part of the feed-forward balanced component. In G_A, the controller contributes only $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$, which attaches the ring asset triangle to the feed-forward component without adding a second odd cycle inside the ring block.

D Gadget Algebra and Error Bounds

We use the encoding $y_v = 2 - p_v$. For a writer with collateral O, debt U, and parameters C, D, ε, the equation is

$$x = \Pi\left(\frac{2D - 2C}{\varepsilon} - \frac{D}{\varepsilon} y_U + \frac{C}{\varepsilon} y_O\right).$$

If it is the unique writer of O, then $y_O = x$. If $C/\varepsilon < 1$, the right-hand side is a contraction in y_O, and any residual δ gives an output error at most $\delta/(1 - C/\varepsilon)$ up to the harmless constant accounting for price and liquidation residuals.

For the NOT gadget, $C = 1, D = 2, \varepsilon = 3$. Hence

$$y_O = \Pi\left(\frac{2 - 2y_U + y_O}{3}\right),$$

whose unique fixed point is $1 - y_U$. The self coefficient is $1/3$, so a conservative error bound is $B_\neg = 3$.

For the average gadget, first maintain complement wires $\bar{U}, \bar{W}$ by NOT gadgets. Create an average asset A with two co-writers, both with collateral A, each contributing $\eta = 1/2$ units of output price drop. Writer 1 reads $\bar{U}$, writer 2 reads $\bar{W}$, and both use $C = 1, D = 2, \varepsilon = 3$. Since $y_{\bar{U}} = 1 - y_U$, writer 1 satisfies

$$x_1 = \frac{2}{3} y_U + \frac{1}{3} y_A,$$

and similarly $x_2 = \frac{2}{3} y_W + \frac{1}{3} y_A$. The output equation is $y_A = (x_1 + x_2)/2$, so $y_A = (y_U + y_W)/2$. The self coefficient is again bounded away from one, and the error is a constant multiple of δ.

The final gain-2 clipped writer reads $\bar{A}$ and uses $C = 1, D = 2, \varepsilon = 2$. Since $p_{\bar{A}} = 2 - (1 - y_A) = 1 + y_A$, the writer equation becomes

$$y_O = \Pi \left(y_A + \frac{1}{2} y_O \right).$$

If $y_A \leq 1/2$, the unique fixed point is $2y_A$; if $y_A > 1/2$, the unique fixed point is 1. Thus the output is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. All self coefficients are at most 1/2 or 1/3 inside the macro, so the total error bound B_+ is an absolute constant.

The close-factor bound can be met by scaling the output impact. For a writer with desired output contribution η, take collateral amount parameter $C = 1$, liquidation amount $\alpha = \varphi/2$, and slope $s_O = 2\eta/\varphi$. Then $s_O \alpha = \eta$ and $\alpha \leq \varphi C$.

E Source Problem and Fan-Out

The restricted source problem in Lemma 7 is taken as a black box. The load-bearing citation is the full version of Budish et al. [6, Appendix D, Definition 10 and Remark 7], which gives a self-contained reduction to simplified circuits containing only SUM and NOT gates, even with bounded fan-out. Filos-Ratsikas et al. [5] and Rubinstein [18] provide related restricted-gate and generalized-circuit hardness background. Our construction does not require bounded fan-out because debt-leg reads are non-invasive.

The unbalanced-source promise used in Lemma 8 is not a target-side graph tag. It is obtained before translation by adding the two gates $a = 1 - s$ and $b = T_{[0,1]}(s + a)$ through an original source wire s. This creates a signed source cycle $s - a - b - s$ with one NOT edge. Lemma 10 then maps that source frustration to an odd collateral-debt cycle in the constructed borrowing graph.

F Subtraction-Free Structure of the Balanced Side

The sign-balanced map in Theorem 1 has the special form

$$y = \Pi_{[0,u]}(Ay + c), \qquad A \geq 0.$$

It is therefore monotone and subtraction-free: it can form nonnegative weighted sums and clip against constants, but it has no direct primitive for comparing two variables. Boolean monotone fragments such as AND/OR have easy least fixed points by finite Kleene iteration, while large-domain Tarski hardness constructions typically use min/max or other variable-comparison operations. Implementing such operations appears to require an anti-monotone dependence, which is precisely what the bipartite sign transform rules out.

LCP location. Writing $B = I - A$, the fixed-point condition is the box complementarity system

$$\begin{aligned} y_i = 0 &\Rightarrow (By + d)_i \geq 0, \\ 0 < y_i < u_i &\Rightarrow (By + d)_i = 0, \\ y_i = u_i &\Rightarrow (By + d)_i \leq 0, \qquad d = -c. \end{aligned}$$

Because $A \geq 0$, B is always a Z-matrix, and the equivalences

$$\begin{aligned} B \text{ is a } P\text{-matrix} &\iff B \text{ is a nonsingular } M\text{-matrix} \\ &\iff \rho(A) < 1 \end{aligned}$$

locate the regimes: small impact is an M-matrix box LCP, while large impact has a Z-matrix base that need not be a P-matrix. The unbounded Z-matrix LCP is polynomial for every Z-matrix; the obstruction here is the upper bound, whose standard-LCP embedding introduces the positive block shown in Appendix A.

Lemma 24 (extremal interior blocks are non-expansive). *Let y^* be the least fixed point of $T(y) = \Pi_{[0,u]}(Ay + c)$ with $A \geq 0$, and let $S = \{i : 0 < y_i^* < u_i\}$. Then $\rho(A_{SS}) \leq 1$. The same holds, by the dual argument, for the interior block of the greatest fixed point.*

Proof. Suppose $\rho(A_{SS}) > 1$. Since $A_{SS} \geq 0$, Perron–Frobenius gives an eigenvector $v \geq 0$, $v \neq 0$, with $A_{SS}v = \lambda v$ and $\lambda > 1$. Define $\tilde{y}_S = y_S^* - \varepsilon v$ and $\tilde{y}_{\bar{S}} = y_{\bar{S}}^*$. For $\varepsilon > 0$ small enough, $\tilde{y}_S \in (0, u_S)$, so the box projection is inactive on S and

$$T_S(\tilde{y}) = (A\tilde{y} + c)_S = y_S^* - \varepsilon A_{SS} v = y_S^* - \varepsilon \lambda v \leq \tilde{y}_S.$$

For a lower-boundary coordinate i, monotonicity and $\tilde{y} \leq y^*$ give $T_i(\tilde{y}) \leq T_i(y^*) = 0 = \tilde{y}_i$. For an upper-boundary coordinate, projection gives $T_i(\tilde{y}) \leq u_i = \tilde{y}_i$. Hence $T(\tilde{y}) \leq \tilde{y}$, so $\tilde{y}$ is a prefixed point, while $\tilde{y} < y^*$. By Knaster–Tarski, the least fixed point is below every prefixed point, contradiction. The greatest-fixed-point statement follows by pushing upward by εv and using the dual postfixed-point characterization.

Why exact active-set methods are delicate. Along the bottom Kleene sequence each coordinate has a monotone phase structure: it can move only from the lower boundary toward the interior and from the interior toward the upper boundary, never in the opposite direction. We do not assert a finite phase-transition bound, since a coordinate may approach its upper bound only asymptotically. Within a fixed active set the dynamics is an affine recursion $y_S^{t+1} = A_{SS}y_S^t + g$, and exact event-driven algorithms appear delicate because first-threshold times reduce to sign questions for matrix-power sequences. Moreover, for $\rho(A) \geq 1$ the box LCP can have several solutions, obstructing the unique-solution structure of UEOPL.

An EOPL-style path viewpoint. The monotone grid iteration $y^{t+1} = T_\eta(y^t)$ with potential $\phi(y) = \mathbf{1}^\top y/\eta$ is a natural path-following viewpoint: T_η is isotone and rounds down, so the potential strictly increases off fixed points and is polynomially bounded. Turning this into an EOPL reduction would require a polynomial predecessor selection for the monotone Kleene path; the forward map is single-valued while predecessors need not be. Since Theorem 3 gives an exact polynomial-time algorithm for affine subtraction-free maps, this viewpoint is recorded only as structural context for grid-based monotone searches.

Complexity of the mixed-sign case (settled). The Subtraction-Free Box Fixed Point problem is in P for arbitrary drift by Theorem 3: for rational $A \geq 0$, arbitrary rational c, and $u > 0$, the least fixed point of $y = \Pi_{[0,u]}(Ay + c)$ is computable in polynomial time. The algorithm combines Lemma 2 (no ramping interior at the least fixed point) with a monotone activation loop that reduces the problem to active blocks whose least fixed point is strictly positive, each solved by an exact from-above cascade; the difficulty is thus combinatorial and algebraic rather than Diophantine. The orbit-following and bounded-magnitude statements recorded below are method-level refinements around this exact algorithm, not its foundation; in particular, whether the one-pass *symmetric* cascade always terminates is a separate method-level question that the activation algorithm sidesteps.

G Subtraction-Free Box Fixed Points: Algorithmic Refinements

We record complementary algorithmic refinements for the balanced map $T(y) = \Pi_{[0,u]}(Ay+c)$, $A \geq 0$, of Theorem 1 and Appendices A and F. The main unconditional polynomial-time result is Theorem 3 for arbitrary drift; Theorem 2 is its nonnegative-drift special case. This appendix should be read as method-level support for orbit-following, bounded-magnitude approximation, and spectrally separated exact computation. Let y^* be the least fixed point, L the input bit length, and $U = \max_i u_i$. A point $y \in [0, u]$ is a *δ-approximate fixed point* if $\|T(y) - y\|_\infty \leq \delta$, the residual notion in which δ-clearing is posed. Throughout we use the stronger interior fact of Lemma 2: at y^*, the strictly interior block $S^* = \{i : 0 < y_i^* < u_i\}$ satisfies $\rho(A_{S^*S^*}) < 1$.

SCC-DAG decomposition. Form the dependency digraph with arc $i \to j$ iff $A_{ij} \neq 0$. Let $C_1, \ldots, C_m$ be its strongly connected components, ordered so that whenever C_b reads C_a we have $a < b$.

Lemma 25 (topological resolution). *$y^*_{C_a}$ is the least fixed point of*

$$T_a(z) = \Pi_{[0,u_{C_a}]}(A_{C_aC_a}z + g_a), \qquad g_a = c_{C_a} + \sum_{b<a} A_{C_aC_b}y^*_{C_b},$$

where g_a is fixed by upstream blocks.

Proof. C_a reads only C_a and components it depends on, among $C_1, \dots, C_{a-1}$. Thus $y^*_{C_a}$ is a fixed point of T_a. Minimality follows by induction, since the global bottom Kleene orbit restricted to C_a is the orbit of T_a once upstream has converged.

A Jordan block of A at the unit circle can arise only from coupling *between* critical SCCs, and is linearized away by resolving each upstream critical SCC to its constant limit. Within each C_a, the matrix $A_{C_aC_a}$ is irreducible, so its peripheral spectrum is ρ times roots of unity and semisimple.

Lemma 26 (residual locality). *For any $y \in [0, u]$ assembled block by block in topological order,*

$$\|T(y) - y\|_\infty = \max_a \|T_a(y_{C_a}) - y_{C_a}\|_\infty,$$

with each T_a using the stored upstream values. Hence resolving every block's submap to residual at most δ yields global residual at most δ.

Proof. For $i \in C_a$, $(Ay + c)_i = (A_{C_aC_a}y_{C_a} + g_a)_i$ with g_a formed from the stored upstream values, so $T(y)_i = T_a(y_{C_a})_i$.

Remark 5. Lemma 26 concerns the residual $\|T(y) - y\|$, the clearing accuracy of interest. Approximating y^* in value would require a Lipschitz propagation bound and per-block accuracy depending on downstream gains; we do not need it.

Per-block taxonomy. It suffices to solve one irreducible block $z = \Pi_{[0,u_S]}(Bz+g)$, $B = A_{SS} \geq 0$, with $r = \rho(B)$.

Lemma 27 (LP detects divergence). *For any $B \geq 0$, the polyhedron $P = \{z \geq 0 : (I - B)z \geq g\}$ is closed under coordinate-wise minimum. If $P \neq \emptyset$, it has a least element, returned by minimizing $\mathbf{1}^\top z$ over P. Moreover, $P = \emptyset$ iff the lower-bounded relaxation $z = \max(Bz + g, 0)$ diverges to $+\infty$.*

Lemma 28 (subcritical upper bound). *If the least element $z^\star$ of P exists with $z^\star \leq u_S$, then $z^\star$ is the exact box fixed point, computed in $\mathrm{poly}(|S|, L)$ time by the LP of Lemma 27. In particular, when $r < 1$ and the residual drift is nonnegative, the unclipped solve $\hat{z} = (I - B)^{-1}g$ is an upper bound on the box least fixed point and is exact if $\hat{z} \leq u_S$. If $\hat{z} \not\leq u_S$, the coordinates with $\hat{z}_i > u_i$ need not be exactly the true clamp set; they may strictly over-cap and must be followed by a verification/repair step rather than accepted as final.*

Proof. $z^\star$ solves the lower-bounded relaxation, whose map dominates T on $[0, u_S]$. Hence the box least fixed point is at most $z^\star$. If $z^\star \le u_S$, the box projection is inactive at $z^\star$, so $z^\star$ is itself a box fixed point and therefore equals the box least fixed point. For $r < 1$ and $g \ge 0$, the unclipped Kleene orbit increases to $\hat{z} = (I - B)^{-1} g$, while the clipped orbit is coordinate-wise no larger. Thus $\hat{z}$ is an upper bound and is exact if it lies inside the box. The over-cap warning is witnessed by

$$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad g = (0, 10), \qquad u = (5, 1),$$

for which $\hat{z} = (10, 10)$, but the box fixed point is $(1, 1)$. Hence the first coordinate is falsely over-threshold in the relaxation.

Remark 6 (relaxation is an upper bound). P drops the box, so its least element dominates y^*_S. Thus "the relaxation diverges" does not imply that the box system diverges to u: a supercritical block can settle finitely when a coordinate clips to 0, breaking the feedback. This is why each block is resolved inside its own box.

Lemma 29 (ramp step counts). *For an irreducible block with $r \ge 1$ whose lower-bounded LP is infeasible, bottom Kleene reaches its first clamp event in $O(U/\bar{\alpha}) \le O(U/\delta)$ iterations when $r = 1$, where $\bar{\alpha}$ is the asymptotic per-step drift envelope over the peripheral period, unless $\bar{\alpha} \le \delta$, in which case the current iterate is already a δ-approximate fixed point; and in $O(\log U/\log r)$ iterations when $r > 1$. Each block contributes at most $|S|$ clamp events.*

Proof. The interior increment is $d^{(t)} = B^t g$. For $r = 1$, the semisimple peripheral spectrum gives linear growth of the partial sums along the Perron direction, with a bounded periodic fluctuation when the block is imprimitive. Taking $\bar{\alpha}$ as the maximum drift over one peripheral period, the position advances at rate at most a constant-period oscillation around $t\bar{\alpha}\hat{v}$, so a cap is reached after $O(U/\bar{\alpha})$ steps; if this envelope is at most δ, the residual is already at most δ. For $r > 1$, the Perron mode dominates and the cap U is reached after $O(\log U/\log r)$ steps.

Approximation, scoped.

Proposition 7 (δ-approximate least fixed point). *Resolving blocks in topological order, using Lemma 28 for LP-bounded exact blocks and verified subcritical relaxation/repair steps, and using bottom Kleene with Lemma 29 for ramping blocks, returns a δ-approximate fixed point in time*

$$O\big(n^2\,(nU/\delta + n\log U/\log r_{\min})\big) = \mathrm{poly}(n, L, \|A\|_\infty, U, 1/\delta).$$

Consequently, for normalized, polynomially bounded instances with $U, \|A\|_\infty = \mathrm{poly}(n, L)$ and $\delta \ge 1/\mathrm{poly}(n, L)$, the running time is polynomial. For arbitrary $A \ge 0, c, u, \delta$, the algorithm is pseudo-polynomial: polynomial in $n, L, \|A\|_\infty, U, 1/\delta$.

Proof. By Lemma 26, the global residual is the maximum per-block residual. LP-bounded blocks have residual 0, and subcritical over-caps are checked and repaired before a block is accepted. A ramping block either has increment $\bar{\alpha} \le \delta$ or reaches a clamp within the count of Lemma 29, after which it is re-decomposed. Iterating leaves every block with residual at most δ. The time bound follows from at most $|S|$ accepted clamp events per block over at most n blocks, each preceded by $O(U/\delta)$ or $O(\log U/\log r)$ iterations of cost $O(n^2)$; the repair steps are included in the method-level accounting.

Corollary 3 (approximate clearing for balanced large-impact markets). *In the transformed system of Lemma 1, $y = (q, z)$ has box upper bounds M_k on prices and 1 on liquidations, and A has off-diagonal entries $s_k\alpha_i$ in the price block and $\theta_i D_i/\varepsilon_i, C_i/\varepsilon_i$ in the liquidation block. Hence $U = \max_k M_k$ and $\|A\|_\infty$ may be exponentially large binary rationals. For instances in which the price caps M_k, impact products $s_k\alpha_i$, and inverse smoothing widths $\theta_i D_i/\varepsilon_i, C_i/\varepsilon_i$ are polynomially bounded, a δ-clearing is computable in polynomial time for inverse-polynomial δ. In general the algorithm is pseudo-polynomial in these magnitudes and $1/\delta$.*

Exact solution under spectral separation.

Definition 2 (γ-spectral separation). *An instance is γ-*spectrally separated *if every irreducible block $B = A_{SS}$ that arises during the full recursive SCC-DAG resolution, including every residual subblock created after clamping coordinates to 0 or u, satisfies $\rho(B) \le 1 - \gamma$ or $\rho(B) \ge 1 + \gamma$, unless the lower-bounded relaxation LP for B is feasible with least element at most u_S, so the block is resolved exactly by the LP with no ramp.*

Theorem 10 (exact polynomial-time solution). *If an instance is γ-spectrally separated with $\gamma \ge 1/\mathrm{poly}(n, L)$, then the exact least fixed point y^* is computable in $\mathrm{poly}(n, L)$ time, even when the box bounds are exponentially large. No tie-breaking hypothesis is needed.*

Proof. Every block encountered by the recursive algorithm is subcritical, LP-bounded, or supercritical with $r \ge 1+\gamma$; none is a critical or barely-supercritical ramp. Subcritical and LP-bounded blocks are exact in polynomial time. A supercritical block reaches its next clamp, or is driven to 0 by lower clipping, within $O(\log U/\log(1+\gamma)) = O(L/\gamma) = \mathrm{poly}(n, L)$ bottom-Kleene iterations. After the clamp, the algorithm re-decomposes the residual subblocks, which are covered again by Definition 2. Thus each of the at most $|S|$ clamp events in the recursive resolution has the same polynomial bound. Once no clamp remains, the interior residual is obtained by one exact linear solve. Simultaneous boundary hits are handled by the clip, so no separation of hitting times is needed.

The critical event-jump. For orbit-following methods, the delicate subcase is an irreducible block with $\rho = 1$ and positive drift ramping to its box. The algebraic cascade of Theorem 2 (nonnegative drift) and the activation algorithm of Theorem 3 (arbitrary drift) bypass this timing problem; the event-jump below remains useful as a certified description of the trajectory and as a method-level refinement.

Proposition 8 (null-vector ramp event, proven special case). *Let $B \geq 0$ be irreducible with $\rho(B) = 1$, constant input g, and suppose the block ramps, i.e., the LP of Lemma 27 is infeasible. Let $v > 0$ be the positive right null vector of $B - I$. Then the bottom Kleene orbit advances along v, and, when B is primitive or the headroom ratios are separated by $\min_{i \neq j} |u_i/v_i - u_j/v_j| \geq 2^{-\mathrm{poly}(n,L)}$, the first coordinate to reach its cap is $j = \arg\min_i u_i/v_i$, independently of the drift magnitude. Clamping j and recursing terminates in at most $|S|$ events, each a null-vector plus a ratio test, in $\mathrm{poly}(|S|, L)$ time with no dependence on U or the drift.*

Status. An unconditional orbit-following event-jump still requires period-h phase alignment for near-tie boundary events, rational bit-complexity of the null-vector and ratio computations, and the post-clamp residual when the reduced block is supercritical. These remain empirically passing but are not used to support Theorem 2 or Theorem 3.

Refinement of the complexity map.

Regime (smooth model)	**Result**
sign-balanced	in CLS
sign-balanced + small impact ($\rho(A) < 1$)	P
sign-balanced + nonnegative drift	P (Thm. 2)
sign-balanced + γ-spectral separation	exact P (Thm. 10)
sign-balanced, bounded caps/impact/$1/\varepsilon$, inverse-poly δ	poly-time δ-approx (Prop. 7)
sign-balanced + arbitrary drift	P (Theorem 3); interior subcritical (Lemma 2)
non-bipartite, large impact	PPAD-hard